Roll No. .....

PAPER ID—21133

B.Tech. (Computer Science and Engineering (Artificial Intelligence and Machine Learning)

EXAMINATION, 2024

(Fourth Semester)

DESIGN AND ANALYSIS OF ALGORITHMS

Time : 3 *Hours* *Maximum Marks :* 70

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Unit. Q. No. **1** is compulsory. All questions carry equal marks.

1. (i) What is time complexity of an algorithm ?

(ii) Explain divide and conquer technique.

(iii) Write merits and demerits of dynamic programming.

(iv) Compare Big Oh and Big Omega notations.

(v) Write a note on Hamiltonian cycle.

(vi) Explain NP hard graph.

(vii) What for graph coloring is used ?

7×2=14

Unit I

2. Explain the relevance of sets and disjoint sets in analysis and design of algorithm. Also discuss graphs. **14**

3. What is binary Search algorithm ? Write and explain its steps. Explain by considering a suitable example. **14**

Unit II

4. What is a greedy technique ? Solve the travelling salesmen problem by using greedy method. **14**

5. Write notes on the following : 7×2=14

(a) Minimum Spanning Tree

(b) Job Sequencing with deadlines.

Unit III

6. Explain 8-Queens problem. Establish lower bounds for sorting by comparison of keys (both average and worst case). 14

7. What is back tracking ? Explain eight queens problem by using back tracking. 14

Unit IV

8. Introduce the concept of NP hard and NP Complete problem ? Explain with examples. 14

9. Write and explain the Cook's theorem. 14